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| United States Patent | 12409848 |
| Kind Code | B2 |
| Date of Patent | September 09, 2025 |
| Inventor(s) | Nakabayashi; Shunichi et al. |

## Vehicle information notification device


### Abstract

This vehicle information notification device includes: a pair of left and right grips which are formed on a steering handle; and a vibration device (which is disposed on at least one of the pair of left and right grips to notify a driver of predetermined information, wherein a plurality of the vibration devices are provided on one of the left and right grips. Both of the pair of left and right grips are respectively provided with a plurality of vibration devices.



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| **Appl. No.:** | **18/036906** |
| **Filed (or PCT Filed):** | **October 18, 2021** |
| **PCT No.:** | **PCT/JP2021/038476** |
| **PCT Pub. No.:** | **WO2022/107529** |
| **PCT Pub. Date:** | May 27, 2022 |

### Prior Publication Data

| **Document Identifier** | **Publication Date** |
|---|---|
| US 20230415765 A1 | Dec. 28, 2023 |

### Foreign Application Priority Data

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|---|---|---|
| JP | 2020-192290 | Nov. 19, 2020 |

## Publication Classification

**Int. Cl.:** **B60W50/16** (20200101); **B60W10/18** (20120101); **B60W10/20** (20060101); **B60W50/14** (20200101); **B62J50/22** (20200101)

**U.S. Cl.:**

CPC **B60W50/16** (20130101); **B60W10/18** (20130101); **B60W10/20** (20130101); **B62J50/22** (20200201); B60W2050/146 (20130101)

## Field of Classification Search

**CPC:** B60T (8/1706); B60W (10/18); B60W (10/20); B60W (2050/146); B60W (2300/36); B60W (50/14); B60W (50/16); B62D (1/14); B62J (27/00); B62J (45/41); B62J (50/22); B62J (6/24); B62K (21/10); B62K (21/26)

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## Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a vehicle information notification device.

(2) The present application claims priority based on Japanese Patent Application No. 2020-192290 filed on Nov. 19, 2020, the contents of which are incorporated herein by reference.

BACKGROUND ART

(3) Conventionally, it is known that a tactile stimulus device provided on a rim of a steering wheel or the like gives information to a driver through the tactile sense during automatic driving of a vehicle (for example, see Patent Document 1).

RELATED ART DOCUMENT

Patent Document

(4) Patent Document 1: Japanese Unexamined Patent Application, First Publication No. 2018-181269

SUMMARY

Problems to be Solved by the Invention

(5) Since the above-described tactile information notification is performed on the part that the driver constantly touches while driving the vehicle compared to the visual information notification, it is possible to reliably notify the driver of information.

(6) However, the amount of notification information is limited in a configuration that simply notifies only based on the presence or absence of tactile stimulation. Therefore, there is a demand for a configuration capable of notifying the driver of more information and increasing the types of information to be notified.

(7) Here, an object of the present invention is to notify a driver of more information and to increase the types of information to be notified in a vehicle information notification device that notifies a driver of information by tactile sense.

Means for Solving the Problem

(8) As a solution to the above problems, a vehicle information notification device of the present invention includes: a pair of left and right grips which are formed on a steering handle; and a vibration device which is disposed on at least one of the pair of left and right grips to notify a driver of predetermined information. A plurality of the vibration devices are provided on one of the left and right grips.

(9) According to this configuration, the vibration device for notifying the driver of information is disposed on at least one of the pair of left and right grips of the steering handle. The plurality of vibration devices are provided on one of the left and right grips. Accordingly, it is possible to combine not only the presence or absence of vibration by the vibration device but also the difference in which of the plurality of vibration devices to be operated. Accordingly, it is possible to notify the driver of more information and to increase the types of information to be notified.

(10) In addition, in the present invention, an operation intervention function is provided to intervene in an operation performed by the driver, and the vibration device is vibrated while being interlocked with the operation intervention function.

(11) According to this configuration, the vibration device is vibrated while being interlocked with the operation intervention function for the driver's operation. Accordingly, it is possible to notify the driver of the operation of the operation intervention function by the vibration of the vibration device.
(12) In addition, in the present invention, in a standby state in which a system of the operation intervention function is activated and the operation intervention function is not operated yet, the vibration device is vibrated as below. That is, the vibration device is vibrated in a vibration mode different from that when the operation intervention function is operated.
(13) According to this configuration, the vibration device is vibrated in a vibration mode different from that when the operation intervention function is operated in the standby state in which only the system of the operation intervention function is activated. Accordingly, it is possible to notify the driver of the standby state in which the operation intervention function is operable.
(14) In addition, in the present invention, the vibration device is vibrated intermittently at a predetermined first time interval in the standby state. The vibration device is vibrated intermittently at a second time interval shorter than the first time interval or is vibrated continuously without any time interval when the operation intervention function is operated.
(15) According to this configuration, the vibration device is vibrated at a time interval shorter than the time interval in the standby state during the operation of the operation intervention function. Accordingly, it is possible to notify the driver that the operation intervention function has been transitioned from the standby state to the operation state.
(16) In the present invention, both of the pair of left and right grips may be respectively provided with a plurality of vibration devices.
(17) According to this configuration, since the plurality of vibration devices are provided on each of the pair of left and right grips, it is also possible to combine the difference in which of the left and right grips is to be vibrated. Accordingly, it is possible to further notify the driver of more information and to further increase the types of information to be notified.
(18) In the present invention, at least one of a steering intervention function and a braking intervention function may be provided as the operation intervention function.
(19) According to this configuration, at least one of the steering intervention function and the braking intervention function in the operation intervention function is operated. Accordingly, it is possible to notify the driver by the vibration of the vibration device.
(20) In the present invention, the plurality of vibration devices provided on one of the left and right grips may be arranged side by side in an axial direction which is elongated in a vehicle width direction of the grip. The vibration device which is disposed on an outside of the grip in the vehicle width direction in the plurality of vibration devices may be vibrated while being interlocked with the steering intervention function.
(21) According to this configuration, the vibration of the vibration device disposed on the outside of the grip in the vehicle width direction in the plurality of vibration devices is interlocked with the steering intervention function. Accordingly, the steering direction by the steering intervention function can be notified by the vibration of the vibration device disposed on the outside of the grip in the vehicle width direction to be easily understood.
(22) In the present invention, the plurality of vibration devices provided on one of the left and right grips may be arranged side by side in the axial direction which is elongated in the vehicle width direction of the grip. The vibration device which is disposed on an inside of the grip in the vehicle width direction in the plurality of vibration devices may be vibrated while being interlocked with the braking intervention function.
(23) According to this configuration, the vibration of the vibration device disposed on the inside of the grip in the vehicle width direction in the plurality of vibration devices is interlocked with the braking intervention function. Accordingly, the operation of the braking intervention function can be notified by the vibration of the vibration device disposed on the inside of the grip in the vehicle

width direction to be easily understood.
(24) In the present invention, the vibration device may be vibrated at a frequency higher than a frequency of the standby state when the operation intervention function is operated.
(25) According to this configuration, when the operation intervention function is operated, the vibration device is vibrated at a frequency higher than that in the standby state. Accordingly, it is possible to notify the driver that the operation intervention function has been transitioned from the standby state to the operation state.
(26) In the present invention, the vibration device may start vibrating at a timing earlier than a timing of starting the operation of the operation intervention function when the operation intervention function is operated.
(27) According to this configuration, the vibration device starts vibrating at a timing earlier than the start of the operation of the operation intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the operation intervention function in advance.
(28) In the present invention, a display device which displays predetermined information for the driver while being interlocked with a vibration of the vibration device may be provided.
(29) According to this configuration, the display device which is interlocked with the vibration of the vibration device is further provided. Accordingly, it is possible to more reliably notify the driver of information by the vibration of the vibration device and the display of the display device.
Advantage of the Invention
(30) According to the present invention, in the vehicle information notification device that notifies the driver of information by tactile sense, it is possible to notify the driver of more information and to increase the types of information to be notified.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a left side view of a motorcycle of an embodiment of the present invention.
(2) FIG. **2** is a perspective view of the periphery of a handle of the motorcycle when viewed from a driver's line of sight.
(3) FIG. **3** is an explanatory diagram of a steering actuator of the motorcycle.
(4) FIG. **4** is a block diagram of a driving assistance device of the motorcycle.
(5) FIG. **5** is an explanatory diagram of an operation of the driving assistance device in a standby state.
(6) FIG. **6** is a graph showing temporal changes in the operation of various devices in the standby state of the driving assistance device.
(7) FIG. **7** is an explanatory diagram of the operation when a right steering intervention of the driving assistance device is operated.
(8) FIG. **8** is a graph showing temporal changes in the operation of various devices when the right steering intervention of the driving assistance device is operated.
(9) FIG. **9** is an explanatory diagram of the operation when a left steering intervention of the driving assistance device is operated.
(10) FIG. **10** is a graph showing temporal changes in the operation of various devices when a left steering intervention of the driving assistance device is operated.
(11) FIG. **11** is an explanatory diagram of the operation when a braking intervention of the driving assistance device is operated.
(12) FIG. **12** is a graph showing temporal changes in the operation of various devices when the braking intervention of the driving assistance device is operated.
DESCRIPTION OF THE EMBODIMENTS
(13) Hereinafter, an embodiment of the present invention will be described with reference to the

drawings. In the following description, directions such as front, rear, left, and right are the same as the directions of the vehicle described below unless otherwise specified. An arrow FR indicating the front of the vehicle, an arrow LH indicating the left of the vehicle, and an arrow UP indicating the upper side of the vehicle are shown at appropriate locations in the drawings used in the following description.

(14) <Whole Vehicle>

(15) FIG. **1** shows a motorcycle **1** as an example of a vehicle of the embodiment. The motorcycle **1** includes a front wheel (steered wheel) **3** which is steered by a handle **2** and a rear wheel (driving wheel) **4** which is driven by a power unit **20**. The motorcycle **1** is a saddle type vehicle in which a driver straddles a vehicle body. The motorcycle **1** is able to swing (bank) the vehicle body in the left and right direction (roll direction) based on ground contact points of the front and rear wheels **3** and **4**. The vehicle of the embodiment is not limited to a vehicle that turns in a direction of banking a vehicle body like a motorcycle. The vehicle of the embodiment includes a vehicle that turns by steering a steered wheel without banking the vehicle body.

(16) The motorcycle **1** includes a steering system component **10**A having a handle **2** and a front wheel **3**. The steering system component **10**A is supported by a head pipe **6** to be steerable. The head pipe **6** is located at a front end portion of a vehicle body frame **5** that forms the skeleton of the motorcycle **1**. The front wheel **3** is supported by lower end portions of a pair of left and right front forks **10** of the steering system component **10**A. The vehicle body frame **5** is surrounded by a vehicle body cover **12**.

(17) The vehicle body frame **5** includes the head pipe **6**, a pair of left and right main frames **7**, a pair of left and right pivot frames **8**, and a pair of left and right seat frames **9**.

(18) The head pipe **6** steerably supports the steering system component **10**A. The left and right main frames **7** extend rearwardly downward from the head pipe **6**. The left and right pivot frames **8** extend downward from the rear end portions of the left and right main frames **7** respectively. The left and right seat frames **9** extend rearwardly upward from the respective upper portions of the left and right pivot frames **8**.

(19) A pivot shaft **8***a* extending in the vehicle width direction is provided between the left and right pivot frames **8**. The front end portion of the swing arm **11** is supported by the left and right pivot frames **8** through the pivot shaft **8***a* to swing up and down. The rear wheel **4** is supported by the rear end portion of the swing arm **11**. A cushion unit (not shown) serving as a shock absorber is provided between the vehicle body frame **5** and the swing arm **11**.

(20) A fuel tank **18** is supported by the upper portions of the left and right main frames **7**. A seat **19** is supported behind the fuel tank **18** by the left and right seat frames **9**.

(21) The power unit **20** of the motorcycle **1** is supported by the left and right main frames **7** and the left and right pivot frames **8**. An output shaft of the power unit **20** is connected to the rear wheel **4** through a chain transmission mechanism (not shown) so that power can be transmitted.

(22) The power unit **20** integrally includes an engine (internal combustion engine) **13** as a prime mover and a transmission **21** connected to the rear of the engine **13**.

(23) The motorcycle **1** includes a front wheel brake **3**B which brakes the front wheel **3** and a rear wheel brake **4**B which brakes the rear wheel **4**. Each of the front wheel brake **3**B and the rear wheel brake **4**B is a disc brake.

(24) The front wheel brake **3**B and the rear wheel brake **4**B appropriately brake the rotation of the front wheel **3** and the rear wheel **4** by operating a brake lever **43** (see FIG. **2**) and a brake pedal (not shown) which are brake operators. Further, the front wheel brake **3**B and the rear wheel brake **4**B appropriately brake the rotation of the front wheel **3** and the rear wheel **4** by operating a brake actuator **102** (see FIG. **4**) which will be described later.

(25) The motorcycle **1** includes a driving assistance device **70** (see FIG. **4**) that assists the driver's driving operation (in the embodiment, the steering operation for steering the front wheel **3** and the braking operation for braking the front wheel **3** and the rear wheel **4**). The driving assistance device

**70** includes a control device **71** that controls a function of automatically intervening in the driver's driving operation (automatic operation intervention function). The automatic operation intervention function includes an automatic steering intervention function and an automatic braking intervention function. The driving assistance device **70** will be described later.
(26) FIG. **2** shows the periphery of the handle **2** at the front part of the vehicle body when viewed from the driver's line of sight. The upper portions of the left and right front forks **10** are supported by the head pipe **6** through a steering stem **31**. The left and right front forks **10** are telescopic shock absorbers. The steering stem **31** includes a top bridge **32** and a bottom bridge **33** which connect the upper portions of the left and right front forks **10** and a stem shaft (steering shaft) **34** which is inserted through the head pipe **6**. The front part of the vehicle body is covered with a front cowl **27** of the vehicle body cover **12**.
(27) For example, the handle **2** of the motorcycle **1** is separate left and right handles, and includes a pair of left and right handles **36** and **37**. For example, the right handle **36** and the left handle **37** are attached to the upper portions of the left and right front forks **10** below the top bridge **32**, respectively.
(28) The right handle **36** includes a right grip **41**A which is gripped by the driver with his/her right hand. The right grip **41**A includes a cylindrical right grip bar **36**A which has a length in the vehicle width direction and extends linearly and an accelerator sleeve **41** which is externally fitted to the right grip bar **36**A to be rotatable around its axis. The right grip bar **36**A is made of a metal pipe or the like. The accelerator sleeve **41** is configured by fixing a rubber grip on the outer periphery of a synthetic resin body.
(29) The right grip bar **36**A extends inward in the vehicle width direction from the accelerator sleeve **41** and supports a right switch box **42** and a right lever holder **44**. Reference numeral **43** in the drawing denotes a right lever (for example, a front brake lever) that is swingably supported by the right lever holder **44**. An end portion of the right grip bar **36**A on the inside in the vehicle width direction is fixed to the upper portion of the right front fork **10** through a right handle clamp **36**B.
(30) The right grip **41**A includes a right actuator **46** for giving vibration as information notification to the driver's right hand. The right actuator **46** is provided on at least one of the right grip bar **36**A and the accelerator sleeve **41**. The right actuator **46** includes a right inner actuator (vibration device) **47** which is disposed on the inside of the right grip **41**A in the vehicle width direction and a right outer actuator (vibration device) **48** which is disposed on the outside of the right grip **41**A in the vehicle width direction. The right inner actuator **47** and the right outer actuator **48** are arranged side by side in the axial direction which is elongated in the vehicle width direction of the right grip **41**A. For example, the right inner actuator **47** and the right outer actuator **48** are composed of piezoelectric elements or vibrators of other structures.
(31) The left handle **37** includes a left grip **51**A which is gripped by the driver's left hand. The left grip **51**A includes a cylindrical left grip bar **37**A which has a length in the vehicle width direction and extends linearly and a handle grip **51** which is fixedly attached to the left grip bar **37**A. The left grip bar **37**A is made of a metal pipe or the like. The handle grip **51** is a rubber grip made of synthetic resin.
(32) The left grip bar **37**A extends inward in the vehicle width direction from the handle grip **51** and supports a left switch box **52** and a left lever holder **54**. Reference numeral **53** in the drawing denotes a left lever (for example, a clutch lever). An end portion of the left grip bar **37**A on the inside of the vehicle width direction is fixed to the upper portion of the left front fork **10** through a left handle clamp **37**B.
(33) The left grip **51**A includes a left actuator **56** for giving vibration as information notification to the driver's left hand. The left actuator **56** is provided on at least one of the left grip bar **37**A and the handle grip **51**. The left actuator **56** includes a left inner actuator (vibration device) **57** which is disposed on the inside of the left grip **51**A in the vehicle width direction and a left outer actuator (vibration device) **58** which is disposed on the outside of the left grip **51**A in the vehicle width

direction. The left inner actuator **57** and the left outer actuator **58** are arranged side by side in the axial direction which is elongated in the vehicle width direction of the left grip **51**A. For example, the left inner actuator **57** and the left outer actuator **58** are composed of piezoelectric elements or vibrators of other structures similarly to the right inner actuator **47** and the right outer actuator **48**.
(34) A meter device **61** is disposed in front of the front fork **10**. The meter device **61** is supported by the vehicle body frame **5** or the front cowl **27**. The meter device **61** includes a display screen **62** such as a liquid crystal display which displays images of a vehicle speed and an engine rotation speed and an indicator lamp group **63** which is arranged around the display screen **62** and notifies various information.
(35) The indicator lamp group **63** includes a right indicator lamp (display device) **66** which is disposed on the right side of the display screen **62** and a left indicator lamp (display device) **67** which is disposed on the left side of the display screen **62**. The right indicator lamp **66** emits light while being interlocked with the operation of the right inner actuator **47** and the right outer actuator **48**. The left indicator lamp **67** emits light while being interlocked with the operation of the left inner actuator **57** and the left outer actuator **58**. The display screen **62** notifies the driver of predetermined information by displaying a predetermined image. The indicator lamp group **63** notifies the driver of predetermined information by performing predetermined light-emitting display (lighting or blinking).
(36) FIG. **3** shows the periphery of the top bridge **32** when viewed from above in the axial direction of the stem shaft **34**.
(37) A steering input is automatically applied to the steering system component **10**A by a steering assist device **73** separately from the operation of the handle **2** by the driver.
(38) The steering assist device **73** includes a steering actuator **74**, an arm **75**, a connecting rod **76**, and a steering control unit **77**.
(39) The steering actuator **74** includes an electric motor which is a drive source of the automatic steering intervention function. The steering actuator **74** is fixed to, for example, the vehicle body frame **5**. A base end portion of the arm **75** is fixed to a drive shaft **74***a* which is an output shaft of the steering actuator **74** to be rotatable together. One end portion of the connecting rod **76** is swingably connected to the tip portion of the arm **75** through a first connecting pin **78**. The other end portion of the connecting rod **76** is swingably connected to a rod connecting portion **32***a* provided on the top bridge **32** through a second connecting pin **79**.
(40) The operation of the steering actuator **74** is controlled by the steering control unit **77**. The output of the steering actuator **74** (the rotation torque of the drive shaft **74***a*) is transmitted to the top bridge **32** through the arm **75** and the connecting rod **76**. Accordingly, the steering actuator **74** generates a steering torque (assist torque) in the steering system component **10**A.
(41) <Driving Assistance Device>
(42) As shown in FIG. **4**, the driving assistance device **70** includes a control device **71**, various sensors **81**, and various devices **82**. The control device **71** controls the operation of various devices **82** based on the detection information acquired from various sensors **81**.
(43) The control device **71** is composed of, for example, a single or a plurality of Electronic Control Units (ECUs). At least a part of the control device **71** may be realized by cooperation of software and hardware.
(44) The control device **71** includes an engine control unit **85**, a steering control unit **77**, a brake control unit **87**, a display control unit **88**, and a grip vibration control unit **89**.
(45) Various sensors **81** include a steering torque sensor **91**, a steering angle sensor **92**, a vehicle body acceleration sensor **93**, a vehicle speed sensor **94**, a camera device **96**, and a radar device **97**.
(46) The steering torque sensor **91** is, for example, a magnetostrictive torque sensor provided between the handle **2** and the steering system component **10**A other than the handle **2**. The steering torque sensor **91** detects a torsional torque (steering input) input from the handle **2** to the other steering system components **10**A.

(47) The steering angle sensor **92** is, for example, a potentiometer provided on the steering shaft (stem shaft **34**). The steering angle sensor **92** detects the turning angle (steering angle) of the steering shaft with respect to the vehicle body.
(48) The vehicle body acceleration sensor **93** is a 5-axis or 6-axis Inertial Measurement Unit (IMU). The vehicle body acceleration sensor **93** detects the angular velocities of the three axes (roll axis, pitch axis, and yaw axis) of the vehicle body and further can estimate the angle and acceleration from the results.
(49) The vehicle speed sensor **94** detects, for example, the rotation speed of the output shaft of the power unit **20**. The vehicle speed sensor **94** can detect the rotation speed of the rear wheel **4** and further the vehicle speed of the motorcycle **1** from the rotation speed.
(50) The camera device **96** includes a camera using a solid-state imaging device such as CCD or CMOS. The camera device **96**, for example, periodically photographs the surroundings of the motorcycle **1** (for example, front, rear, left, and right) using the camera. The camera device **96** generates image data from the captured image through image processing such as filtering and binarization.
(51) The radar device **97** radiates radio waves such as millimeter waves around the motorcycle **1**. The radar device **97** detects radio waves (reflected waves) reflected by an object around the vehicle. The radar device **97** can detect at least the front, rear, left, and right positions of the object with respect to the motorcycle **1** (distance and orientation with respect to the motorcycle **1**) and the speed.
(52) Information from the camera device **96** and the radar device **97** described above is used to recognize the position, type, speed, and the like of an object in the detection direction. Based on this recognition, driving assistance control, automatic driving control, and the like of the motorcycle **1** are performed.
(53) Various devices **82** include the steering actuator **74**, the brake actuator **102**, the indicator lamp group **63**, and a grip actuator **104**.
(54) The steering actuator **74** generates a steering torque for steering the front wheel **3** independently of the operation of the handle **2** by the driver. The steering actuator **74** may also serve as a steering damper.
(55) The brake actuator **102** operates the front wheel brake **3**B and the rear wheel brake **4**B by supplying a hydraulic pressure to the front wheel brake **3**B and the rear wheel brake **4**B separately from the operation of the brake operator by the driver. The brake actuator **102** may also serve as an Anti-lock Brake System (ABS) control unit. The brake actuator **102** may be connected to a brake line branched from the normal brake circuit.
(56) The indicator lamp group **63** includes a right indicator lamp **66** and a left indicator lamp **67**. The right indicator lamp **66** and the left indicator lamp **67** emit light while being interlocked with the operation of the grip actuator **104**.
(57) The grip actuator **104** includes the right inner actuator **47**, the right outer actuator **48**, the left inner actuator **57**, and the left outer actuator **58**. In the embodiment, the right inner actuator **47** and the left inner actuator **57** are set to have vibration frequencies lower than those of the right outer actuator **48** and the left outer actuator **58**. Additionally, the vibration frequency of each of the actuators **47**, **48**, **57**, and **58** may be variable.
(58) Next, the control device **71** will be described.
(59) The engine control unit **85** controls the output of the engine **13** based on the throttle opening, intake negative pressure, fuel injection amount, valve timing, ignition timing, and the like in the engine **13**. Further, the vehicle speed of the motorcycle **1** is changed according to the crankshaft rotation speed of the engine **13** and the gear ratio of the transmission **21** by controlling the output of the engine **13**.
(60) The steering control unit **77** controls the operation of the steering actuator **74** based on the following signals and information. The signals and information are the steering torque signal

detected by the steering torque sensor **91**, the angular velocity signal detected by the vehicle body acceleration sensor **93**, the vehicle speed signal detected by the vehicle speed sensor **94**, the detection information detected by the camera device **96** and the radar device **97**, and the like. Accordingly, the assist torque is applied to the steering system component **10**A. By this assist torque, the steering of the front wheel **3** which is a steered wheel is assisted. In this way, the steering control unit **77** controls the automatic steering intervention function.
(61) The brake control unit **87** controls the operation of the brake actuator **102** based on the engine output, the vehicle speed signal detected by the vehicle speed sensor **94**, the detection information detected by the camera device **96** and the radar device **97**, and the like. Accordingly, the front wheel brake **3**B and the rear wheel brake **4**B generate an assist braking force. By this assist braking force, the braking of the front wheel **3** and the rear wheel **4** is assisted. In this way, the brake control unit **87** controls the automatic braking intervention function.
(62) The display control unit **88** controls the light emission (lighting or blinking) of the right indicator lamp **66** and the left indicator lamp **67** in accordance with the control of the following functions. The above-described function control includes the control of the automatic steering intervention function by the steering control unit **77** and the control of the automatic braking intervention function by the brake control unit **87**.
(63) The grip vibration control unit **89** controls the operations of the right inner actuator **47**, the right outer actuator **48**, the left inner actuator **57**, and the left outer actuator **58** in accordance with the control of the following functions. The above-described function control includes the control of the automatic steering intervention function by the steering control unit **77** and the control of the automatic braking intervention function by the brake control unit **87**.
(64) The display control unit **88** and the grip vibration control unit **89** are synchronously controlled when the steering actuator **74** and the brake actuator **102** are operated.
(65) The engine control unit **85**, the steering control unit **77**, the brake control unit **87**, the display control unit **88**, and the grip vibration control unit **89** described above all include microcomputers and are configured to communicate with each other.
(66) The operation of the driving assistance device **70** described above will be described.
(67) FIG. **5** shows a standby state in which the system of the driving assistance device **70** is activated and the automatic operation intervention function is not operated yet. In the standby state, the main switch (power supply) of the motorcycle **1** is turned on and the operation switch of the driving assistance device **70** is turned on so that the operation of the automatic operation intervention function is allowed. In the standby state, various sensors **81** monitor various parameters and the like for operating the automatic operation intervention function. Based on this monitoring information, the automatic operation intervention function can be operated immediately.
(68) In the standby state, the right inner actuator **47** of the right grip **41**A and the left inner actuator **57** of the left grip **51**A are vibrated in a predetermined mode. Accordingly, this notifies the driver that the system is activated. Additionally, FIG. **5** and FIGS. **7**, **9**, and **11** below show that those which are operated (vibrated) are shaded among the right inner actuator **47**, the right outer actuator **48**, the left inner actuator **57**, the left outer actuator **58**.
(69) FIG. **6** shows the operation of the grip actuator **104** and the like in the standby state as time goes by.
(70) The vertical axis of the graph of FIG. **6** indicates the operation state (ON (operation) and OFF (non-operation)) of the actuators **47**, **48**, **57**, and **58** of the left and right grips **41**A and **51**A together with the left and right indicator lamps **66** and **67**. The vertical axis further indicates the operation state (ON and OFF) of the steering intervention or the braking intervention. The horizontal axis of the graph indicates the time T (sec).
(71) In the drawing, “RH inside” indicates the right inner actuator **47**, “LH inside” indicates the left inner actuator **57**, “RH outside” indicates the right outer actuator **48**, and “LH outside” indicates

the left outer actuator **58**. Further, "indRH" indicates the right indicator lamp **66** and "indLH" indicates the left indicator lamp **67**. Additionally, even in the graphs of FIGS. **8**, **10**, and **12**, the vertical axis and the horizontal axis are the same as those of the graph shown in FIG. **6**.

(72) In the above-described standby state, the right inner actuator **47** and the left inner actuator **57** are vibrated intermittently at the same timing. In the standby state, the right inner actuator **47** and the left inner actuator **57** are simultaneously vibrated for a predetermined time T**1** (for example, 0.5 to 1 second). Further, the right inner actuator **47** and the left inner actuator **57** repeat the vibration of the time T**1** at the time interval of a predetermined time T**2** (for example, about 2 seconds). In the standby state, the right inner actuator **47** and the left inner actuator **57** are vibrated intermittently at a predetermined low frequency L.

(73) FIG. **7** shows a state in which the driving assistance device **70** operates the automatic steering intervention function from the standby state and is about to turn the course of the motorcycle **1** to the right side.

(74) For example, when the steering control unit **77** of the driving assistance device **70** determines that the motorcycle **1** has moved too far to the left side of the lane, the steering control unit **77** tries to return the course of the motorcycle **1** to the right side.

(75) At this time, the steering control unit **77** steers the front wheel **3** to the left side by the steering actuator **74**. That is, the front wheel **3** is steered in the opposite direction to the side on which the course of the motorcycle **1** is to be changed (reverse steering). Accordingly, the vehicle body of the motorcycle **1** rolls toward the side where the course is to be changed. As a result, the motorcycle **1** can be naturally steered in the course change direction.

(76) In the case of the vehicle that turns by steering the steered wheel without banking the vehicle body, the steered wheel is not steered in the opposite direction, but is steered forward in the course change direction.

(77) When the above-described automatic steering intervention function is operated, only the right outer actuator **48** of the right grip **41**A in the course change direction (right side) of the motorcycle **1** is vibrated in a vibration mode different from that in the standby state. Further, the right indicator lamp **66** of the meter device **61** starts blinking.

(78) Accordingly, the driver can recognize the following points through the tactile sense of the hands holding the left and right grips **41**A and **51**A and the visual sense. That is, the driver can recognize the operation of the automatic steering intervention function and the moving direction of the motorcycle **1**.

(79) FIG. **8** shows the operation mode of each of the actuators **47**, **48**, **57**, and **58** during the operation of the automatic steering intervention function shown in FIG. **7**.

(80) During the operation of the automatic steering intervention function shown in FIG. **7**, only the right outer actuator **48** of the right grip **41**A starts vibrating for a predetermined time T**3** (for example, 0.5 to 1 second). Further, the right outer actuator **48** repeats the vibration of the time T**3** at the time interval of a predetermined time T**4** (for example, 0.5 to 1 second). The time T**4** is shorter than the time T**2** in the standby state. The time T**3** may be equal to or shorter than the time T**1** in the standby state.

(81) That is, the right outer actuator **48** is vibrated intermittently at a period (T**3**+T**4**) shorter than a period (T**1**+T**2**) in the standby state during the above-described operation.

(82) Further, the right outer actuator **48** is vibrated intermittently at a predetermined high frequency H during the above-described operation. The frequency of the high frequency H is higher than that of the low frequency L in the standby state.

(83) The vibration frequencies L and H when the actuators **47**, **48**, **57**, and **58** are vibrated are set in the following frequency band. That is, the vibration frequencies L and H are set in a frequency band that avoids vibration frequencies unique to the motorcycle **1** such as engine vibration and wheel vibration. Accordingly, the vibration of each of the actuators **47**, **48**, **57**, and **58** is difficult to be confused with other vibrations. As a result, it becomes easier for the driver to perceive the

vibrations of the actuators **47**, **48**, **57**, and **58**.
(84) When the right outer actuator **48** is vibrated during the above-described operation, the following operations occur. That is, the right indicator lamp **66** emits light (blinks) while being interlocked (synchronized) with the vibration of the right outer actuator **48**. Additionally, the right indicator lamp **66** may blink in a period having a phase with respect to the vibration period of the right outer actuator **48** or may blink in a different period.
(85) Here, a timing ta at which the right outer actuator **48** starts vibrating and the right indicator lamp **66** starts blinking is set as follows. That is, the timing ta is earlier than a timing tb at which the automatic steering intervention function starts operating by a time td (for example, about 1 second). That is, the vibrating of the right outer actuator **48** and the blinking of the right indicator lamp **66** start before the operation of the automatic steering intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the automatic steering intervention function in advance.
(86) Additionally, the vibrating of the right outer actuator **48** and the blinking of the right indicator lamp **66** stop simultaneously at a timing tc when the operation of the automatic steering intervention function stops.
(87) FIG. **9** shows a state in which the driving assistance device **70** operates the automatic steering intervention function from the standby state and is about to turn the course of the motorcycle **1** to the left side.
(88) For example, when the steering control unit **77** of the driving assistance device **70** determines that the motorcycle **1** has moved too far to the right side of the lane, the steering control unit **77** tries to return the course of the motorcycle **1** to the left side.
(89) At this time, the steering control unit **77** steers the front wheel **3** to the right side by the steering actuator **74**. That is, the front wheel **3** is steered in the opposite direction to the side on which the course of the motorcycle **1** is to be changed (reverse steering). Accordingly, the motorcycle **1** can be naturally steered in the course change direction.
(90) During the operation of the automatic steering intervention function, only the left outer actuator **58** of the left grip **41**A in the course changing direction (left side) of the motorcycle **1** is vibrated in a vibration mode different from that in the standby state. Further, the left indicator lamp **67** of the meter device **61** starts blinking.
(91) Accordingly, the driver can recognize the following points through the tactile sense of the hands holding the left and right grips **41**A and **51**A and the visual sense. That is, the driver can recognize the operation of the automatic steering intervention function and the moving direction of the motorcycle **1**.
(92) FIG. **10** shows the operation mode of each of the actuators **47**, **48**, **57**, and **58** during the operation of the automatic steering intervention function shown in FIG. **9**.
(93) During the operation of the automatic steering intervention function shown in FIG. **9**, only the left outer actuator **58** of the left grip **41**A starts vibrating for a predetermined time T**3** (for example, 0.5 to 1 second). Further, the left outer actuator **58** repeats the vibration of the time T**3** at the time interval of the predetermined time T**4** (for example, 0.5 to 1 second). That is, the left outer actuator **58** is vibrated intermittently at a period (T**3**+T**4**) shorter than a period (T**1**+T**2**) in the standby state during the above-described operation.
(94) Further, the left outer actuator **58** is vibrated intermittently at a predetermined high frequency H during the above-described operation.
(95) When the left outer actuator **58** is vibrated during the above-described operation, the following operations occur. That is, the left indicator lamp **67** also emits light while being interlocked (synchronized) with the vibration of the left outer actuator **58**.
(96) Here, the timing ta at which the left outer actuator **58** starts vibrating and the left indicator lamp **67** starts blinking is set as follows. That is, the timing ta is earlier than a timing tb at which the automatic steering intervention function starts operating by a time td (for example, about 1

second). That is, the vibrating of the left outer actuator **58** and the blinking of the left indicator lamp **67** start before the operation of the automatic steering intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the automatic steering intervention function in advance.
(97) Additionally, the vibrating of the left outer actuator **58** and the blinking of the left indicator lamp **67** stop simultaneously at the timing tc when the operation of the automatic steering intervention function stops.
(98) FIG. **11** shows a state in which the driving assistance device **70** operates the automatic braking intervention function from the standby state and is about to brake the motorcycle **1**.
(99) The automatic braking intervention function of the embodiment vibrates both of the left and right grips **41**A and **51**A since there is no relevance in the left and right direction. The automatic braking intervention function is assumed to induce the driver's bodily behavior and to be urgent. Therefore, all of the actuators **47**, **48**, **57**, and **58** of the left and right grips **41**A and **51**A are vibrated. At this time, each of the actuators **47**, **48**, **57**, and **58** is vibrated continuously without any time interval instead of intermittently vibrating.
(100) Further, both of the right indicator lamp **66** and the left indicator lamp **67** emit light (light up) while being interlocked (synchronized) with the vibration of each of the actuators **47**, **48**, **57**, and **58**. This light emission is not limited to lighting, and may be blinking.
(101) FIG. **12** shows the operation mode of each of the actuators **47**, **48**, **57**, and **58** during the operation of the automatic braking intervention function.
(102) When the automatic braking intervention function is operated by an obstacle or the like in front of the motorcycle **1**, all of the actuators **47**, **48**, **57**, and **58** start vibrating. Further, both of the indicator lamps **66** and **67** start lighting while being interlocked (synchronized) with this vibration. These operations are performed continuously without a time interval during the time T**5** when the automatic braking intervention function is operated.
(103) At this time, the right inner actuator **47** and the left inner actuator **57** are vibrated at a low frequency L. The right outer actuator **48** and the left outer actuator **58** are vibrated at a high frequency H.
(104) Here, a timing te at which each of the actuators **47**, **48**, **57**, and **58** starts vibrating and each of the indicator lamps **66** and **67** starts lighting is set as follows. That is, the timing te is earlier than a timing tf at which the automatic braking intervention function starts operating by a time th (for example, about 1 second). That is, the vibrating of each of the actuators **47**, **48**, **57**, and **58** and the lighting of each of the indicator lamps **66** and **67** are started before the operation of the automatic braking intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the automatic braking intervention function in advance.
(105) Additionally, the vibrating of each of the actuators **47**, **48**, **57**, and **58** and the blinking of each of the indicator lamps **66** and **67** stop simultaneously at the timing tg when the operation of the automatic steering intervention function stops.
(106) As described above, the vehicle information notification device of the above-described embodiment includes the pair of left and right grips **41**A and **51**A formed on the steering handle **2** and the vibration device (the right inner actuator **47**, the right outer actuator **48**, the left inner actuator **57**, and the left outer actuator **58**) disposed on at least one of the pair of left and right grips **41**A and **51**A (both in the embodiment) to notify the driver of predetermined information. The plurality of vibration devices are provided on one of the left and right grips. That is, the right inner actuator **47** and the right outer actuator **48** are provided on the right grip **41**A. The left inner actuator **57** and the left outer actuator **58** are provided on the left grip **51**A.
(107) According to this configuration, the vibration device for notifying the driver of information is disposed on at least one of the pair of left and right grips **41**A and **51**A of the steering handle **2**. The plurality of vibration devices are provided on at least one of the left and right grips. Accordingly, it is possible to combine not only the presence or absence of vibration by the vibration device but

also the difference in which of the plurality of vibration devices are to be operated. Accordingly, it is possible to notify the driver of more information and to increase the types of information to be notified.

(108) Further, in the above-described vehicle information notification device, the plurality of vibration devices are provided on each of both of the pair of left and right grips **41**A and **51**A. Accordingly, it is also possible to combine the difference in which of the left and right grips **41**A and **51**A is to be vibrated. Accordingly, it is possible to further notify the driver of more information and to further increase the types of information to be notified.

(109) Further, the above-described vehicle information notification device has an operation intervention function that can intervene in the operation performed by the driver. Each of the plurality of vibration devices is vibrated while being interlocked with the operation intervention function. Accordingly, it is possible to notify the driver of the operation of the operation intervention function by the vibration of the vibration device.

(110) Further, the above-described vehicle information notification device has at least one of the steering intervention function and the braking intervention function (both in the embodiment) as the operation intervention function. Accordingly, it is possible to notify the driver of the operation of at least one of the steering intervention function and the braking intervention function by the vibration of the vibration device.

(111) Further, in the vehicle information notification device, the plurality of vibration devices (the right inner actuator **47** and the right outer actuator **48** in the right grip **41**A and the left inner actuator **57** and the left outer actuator **58** in the left grip **51**A) provided on one of the left and right grips are arranged side by side in the axial direction which is elongated in the vehicle width direction of each of the grips **41**A and **51**A. In the plurality of vibration devices provided on each of the grips **41**A and **51**A, the vibration device (the right outer actuator **48** and the left outer actuator **58**) which is disposed on the outside of each of the grips **41**A and **51**A in the vehicle width direction is vibrated while being interlocked with the steering intervention function. Accordingly, it is possible to notify the steering direction of the steering intervention function by the vibration of the vibration device on the outside of each of the grips **41**A and **51**A in the vehicle width direction to be easily understood.

(112) Further, in the vehicle information notification device, the vibration device (the right inner actuator **47** and the left inner actuator **57**) disposed on the inside of each of the grips **41**A and **51**A in the vehicle width direction in the plurality of vibration devices provided on each of the grips **41**A and **51**A is vibrated while being interlocked with the braking intervention function. Accordingly, it is possible to notify the operation of the braking intervention function by the vibration of the vibration device on the inside of each of the grips **41**A and **51**A in the vehicle width direction to be easily understood.

(113) Further, in the vehicle information notification device, the plurality of vibration devices are vibrated as below in the standby state in which the system of the operation intervention function is activated and the operation intervention function is not operated yet. That is, the plurality of vibration devices are vibrated in a vibration mode different from that when the operation intervention function is operated. Accordingly, it is possible to notify the driver of the standby state in which the system of the operation intervention function is activated and the operation intervention function is operable.

(114) Further, in the vehicle information notification device, the vibration device (the right outer actuator **48** and the left outer actuator **58**) operated when the operation intervention function is operated in the plurality of vibration devices is vibrated as below. That is, the vibration device which is operated when the operation intervention function is operated is vibrated at a frequency H higher than a frequency L of the vibration device (the right inner actuator **47** and the left inner actuator **57**) operated in the standby state. Accordingly, it is possible to notify the driver that the operation intervention function has been transitioned from the standby state to the operation state.

(115) Further, in the vehicle information notification device, the vibration device (the right inner actuator **47** and the left inner actuator **57**) operated in the standby state in the plurality of vibration devices is vibrated intermittently at a predetermined first time interval. The vibration device (each of the actuators **47**, **48**, **57**, and **58**) operated when the operation intervention function is operated is vibrated intermittently at a second time interval shorter than the first time interval or is vibrated continuously without any time interval. Even in this configuration, it is possible to notify the driver that the operation intervention function has transitioned from the standby state to the operation state.
(116) Further, in the vehicle information notification device, when the operation intervention function is operated, the vibration device (each of the actuators **47**, **48**, **57**, and **58**) starts vibrating at the timings ta and te earlier than the timings tb and tf of the start of the operation of the operation intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the operation intervention function in advance.
(117) Further, in the vehicle information notification device in, the display device (the right indicator lamp **66** and the left indicator lamp **67**) is further provided to display predetermined information for the driver while being interlocked with the vibration of the vibration device (each of the actuators **47**, **48**, **57**, and **58**). Accordingly, it is possible to more reliably notify the driver of information by the vibration of the vibration device and the display of the display device.
(118) Additionally, the present invention is not limited to the above-described embodiment. For example, the driving assistance device **70** has the steering intervention function and the braking intervention function as the operation intervention function, but the present invention is not limited thereto. That is, the driving assistance device **70** may have only one of the steering intervention function and the braking intervention function as the operation intervention function.
(119) In the embodiment, an example of notifying operation information of the device related to driving assistance is shown, but the present invention is not limited thereto. That is, the notification may be, for example, information from a navigation system or ITS (Intelligent Transport Systems).
(120) The present invention is not limited to a configuration in which the plurality of vibration devices are provided on both of the left and right grips. That is, the present invention may be a configuration in which the plurality of vibration devices are provided only in one of the left and right grips. Further, one of the left and right grips may include a single vibration device or may not include any vibration device.
(121) The handle according to the present invention is not limited to left and right separate handles. That is, a left and right integrated bar handle may be used and the handle may not be a bar type handle.
(122) The saddle type vehicle includes all vehicles in which the driver straddles the vehicle body. That is, the saddle type vehicle includes not only motorcycles (including motorized bicycles and scooter-type vehicles), but also vehicles with three wheels (including vehicles with one front wheel and two rear wheels as well as vehicles with two front wheels and one rear wheel) or four wheels. Further, the saddle type vehicle also includes vehicles whose prime mover includes an electric motor. Further, the present invention is not limited to a saddle type vehicle and can be applied to general vehicles having left and right grips on steering handles.
(123) Then, the configuration of the above-described embodiment is an example of the present invention. That is, various modifications can be made without departing from the gist of the present invention, such as replacing the constituent elements of the embodiments with well-known constituent elements.
BRIEF DESCRIPTION OF THE REFERENCE SYMBOLS
(124) **1**: Motorcycle (saddle type vehicle) **2**: Handle **41**A: Right grip (grip) **47**: Right inner actuator (vibration device) **48**: Right outer actuator (vibration device) **51**A: Left grip (grip) **57**: Left inner actuator (vibration device) **58**: Left outer actuator (vibration device) **66**: Right indicator lamp (display device) **67**: Left indicator lamp (display device) L: Low frequency (frequency of standby

state) H: High frequency (frequency higher than standby state) T**2**: (First time interval) T**4**: (Second time interval) tb and tf: Timing to start operation intervention function ta and te: (Timing earlier than start of operation intervention function)

## Claims

1. A vehicle information notification device comprising: a pair of left and right grips which are formed on a steering handle; and a vibration device which is an actuator disposed on at least one of the pair of left and right grips to notify a driver by giving predetermined vibration information, wherein a plurality of vibration devices are provided on one of the left and right grips, wherein an operation intervention function is provided to intervene in an operation performed by the driver based on an object information detected by a camera or a radar, wherein the operation intervention function includes at least one of a steering intervention function that assists the steering handle by applying an assist torque and a breaking intervention function that assist a breaking of vehicle wheel by generating an assist braking force, wherein the vibration device is vibrated while being interlocked with the operation intervention function, wherein the vibration device is vibrated in a vibration mode different from that when the operation intervention function is operated in a standby state in which a system of the operation intervention function is activated and the operation intervention function is not operated yet, and wherein the vibration device is vibrated intermittently at a predetermined first time interval in the standby state and is vibrated intermittently at a second time interval shorter than the first time interval or is vibrated continuously without any time interval when the operation intervention function is operated.
2. The vehicle information notification device according to claim 1, wherein both of the pair of left and right grips are respectively provided with the plurality of vibration devices.
3. The vehicle information notification device according to claim 1, wherein the plurality of vibration devices provided on one of the left and right grips are arranged side by side in an axial direction which is elongated in a vehicle width direction of the grip, and wherein the vibration device which is disposed on an outside of the grip in the vehicle width direction in the plurality of vibration devices is vibrated while being interlocked with the steering intervention function.
4. The vehicle information notification device according to claim 1, wherein the plurality of vibration devices provided on one of the left and right grips are arranged side by side in the axial direction which is elongated in the vehicle width direction of the grip, and wherein the vibration device which is disposed on an inside of the grip in the vehicle width direction in the plurality of vibration devices is vibrated while being interlocked with the braking intervention function.
5. The vehicle information notification device according to claim 1, wherein the vibration device is vibrated at a first frequency higher than a second frequency of the standby state when the operation intervention function is operated.
6. The vehicle information notification device according to claim 1, wherein the vibration device starts vibrating at a timing earlier than a timing of starting the operation of the operation intervention function when the operation intervention function is operated.
7. The vehicle information notification device according to claim 1, further comprising: a display device which displays the predetermined vibration information for the driver while being interlocked with a vibration of the vibration device.
8. The vehicle information notification device according to claim 1, wherein the operation intervention function includes both of the automatic steering intervention function and the automatic braking intervention function, wherein the vibration device is vibrated intermittently at predetermined first time interval in the standby state and is vibrated intermittently at a second time interval shorter than the first time interval when the automatic steering intervention function is operated, and wherein the vibration device is vibrated continuously without any time interval when the automatic braking intervention function is operated.

9. The vehicle information notification device according to claim 1, wherein the vehicle is a motorcycle or the like, and steering intervention function steers the steered wheel in the opposite direction to a side on which a course a vehicle is to be changed, and wherein only one side of the vibration device on the course change direction of the vehicle is vibrated when the steering intervention function is operated.